

Tecnológico Nacional de México

Campus Saltillo

Ingeniería en Sistemas Computacionales

Cómputo en la nube

4.3 Seguridad (hardening) en servidor web Apache 2

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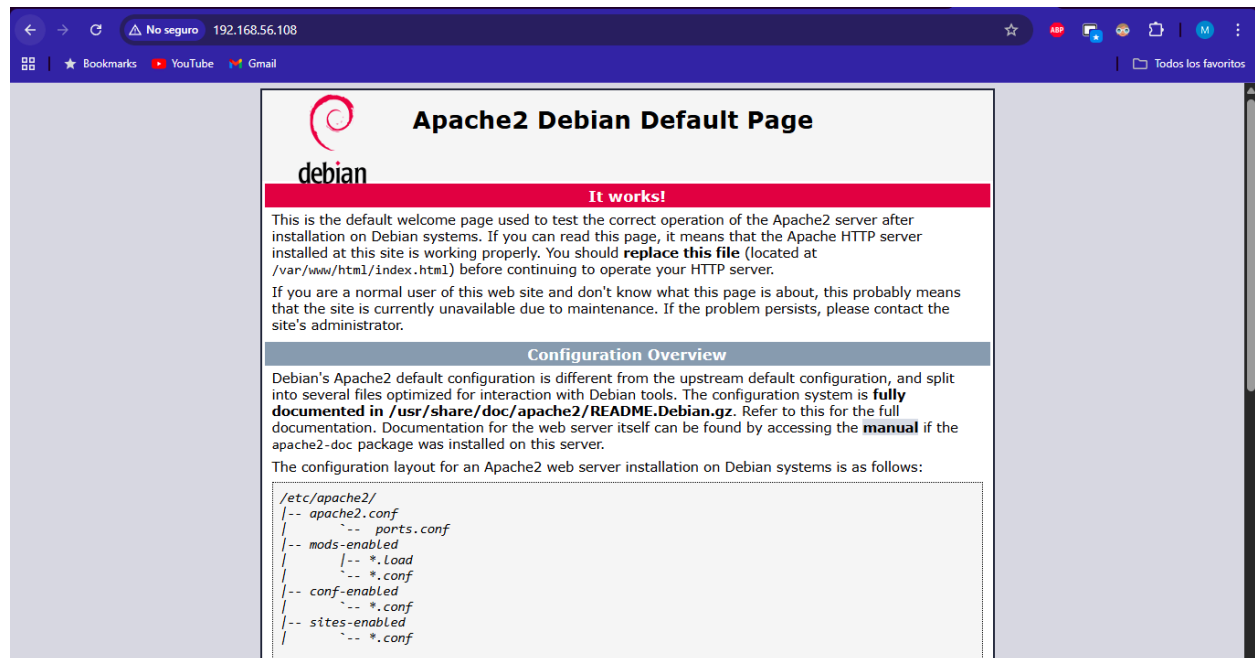
Paso 1: Instalación del servidor Apache

```
Get:7 http://security.debian.org/debian-security trixie-security/main Translation-en [74.3 kB]
Get:8 http://deb.debian.org/debian trixie/main amd64 Packages [9,671 kB]
Get:9 http://deb.debian.org/debian trixie/main Translation-en [6,484 kB]
Fetched 27.3 MB in 11s (2,479 kB/s)
56 packages can be upgraded. Run 'apt list --upgradable' to see them.
Notice: Repository 'http://deb.debian.org/debian trixie InRelease' changed its 'Version' value from '13.3' to '13.4'
Warning: An error occurred during the signature verification. The repository is not updated and the previous index file
will be used. OpenPGP signature verification failed: http://deb.debian.org/debian trixie-updates InRelease: Sub-pro
/usr/bin/sq returned an error code (1), error message is: Verifying signature: Not live until 2026-03-20T07:38:01Z
:38:01Z Verifying signature: Not live until 2026-03-20T07:38:14Z
Warning: Failed to fetch http://deb.debian.org/debian/dists/trixie-updates/InRelease Sub-process /usr/bin/sq return
an error code (1), error message is: Verifying signature: Not live until 2026-03-20T07:38:01Z Verifying s
ture: Not live until 2026-03-20T07:38:14Z
Warning: Some index files failed to download. They have been ignored, or old ones used instead.
Upgrading:
  apache2 apache2-bin apache2-data apache2-utils

Summary:
  Upgrading: 4, Installing: 0, Removing: 0, Not Upgrading: 52
  Download size: 2,011 kB
  Space needed: 0 B / 16.0 GB available

Get:1 http://deb.debian.org/debian trixie/main amd64 apache2 amd64 2.4.66-1~deb13u2 [225 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 apache2-bin amd64 2.4.66-1~deb13u2 [1,409 kB]
Get:3 http://deb.debian.org/debian trixie/main amd64 apache2-data all 2.4.66-1~deb13u2 [160 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 apache2-utils amd64 2.4.66-1~deb13u2 [216 kB]
Fetched 2,011 kB in 0s (4,923 kB/s)
apt-listchanges: Reading changelogs...
(Reading database ... 43336 files and directories currently installed.)
```

Paso 2: Verificar funcionamiento en el navegador



← → ↻ No seguro 192.168.56.108 ☆ APP 📄 📁 📧 M ⋮

📁 ★ Bookmarks 📺 YouTube 📧 Gmail | 📁 Todos los favoritos



Apache2 Debian Default Page

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Debian systems. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Debian's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Debian tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Debian systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
|-- conf-enabled
|   |-- *.conf
|-- sites-enabled
|   |-- *.conf
```

Paso 3: Ocultar información sensible



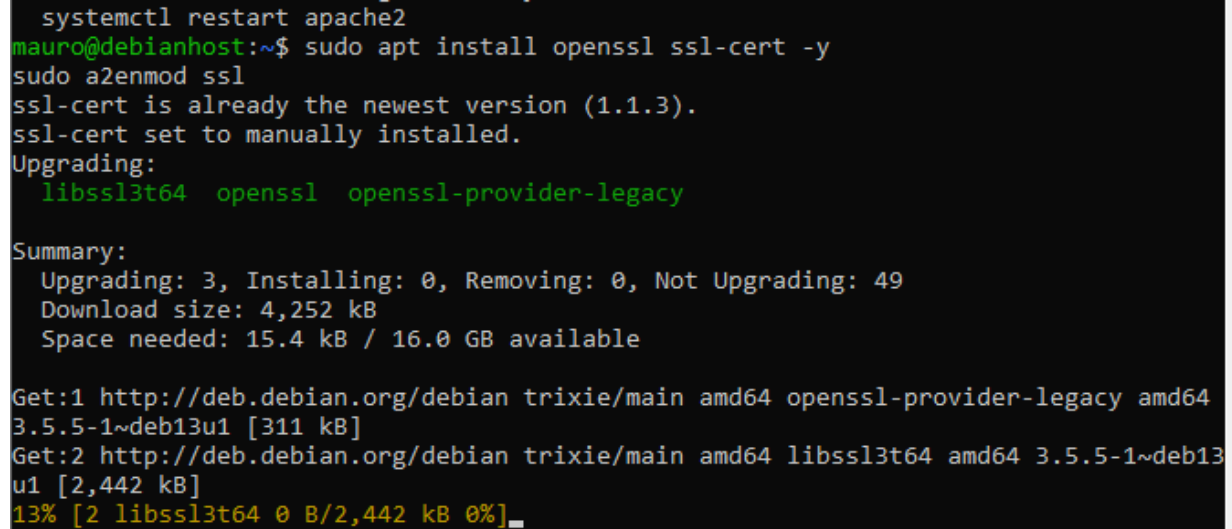
```
mauro@debianhost: ~  
mauro@debianhost:~$ sudo nano /etc/apache2/conf-available/security.conf  
mauro@debianhost:~$ sudo a2enconf security  
do systemctl restart apache2  
nf security already enabled  
mauro@debianhost:~$ _
```

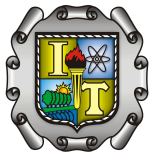
PASO 4 — Eliminar módulos innecesarios:

```
mauro@debianhost:~$ apache2ctl -M  
sudo a2dismod autoindex status proxy proxy_http  
sudo systemctl restart apache2  
-bash: apache2ctl: command not found  
WARNING: The following essential module will be disabled.  
This might result in unexpected behavior and should NOT be done  
unless you know exactly what you are doing!  
autoindex  
  
To continue type in the phrase 'Yes, do as I say!' or retry by passing '-f': Yes, do as I say!  
Module autoindex disabled.  
Module status disabled.  
Module proxy already disabled  
Module proxy_http already disabled  
To activate the new configuration, you need to run:  
systemctl restart apache2  
mauro@debianhost:~$ _
```

Paso 5: Configurar HTTPS (SSL) con certificado autofirmado

5.1 Instalar módulos SSL y herramientas:

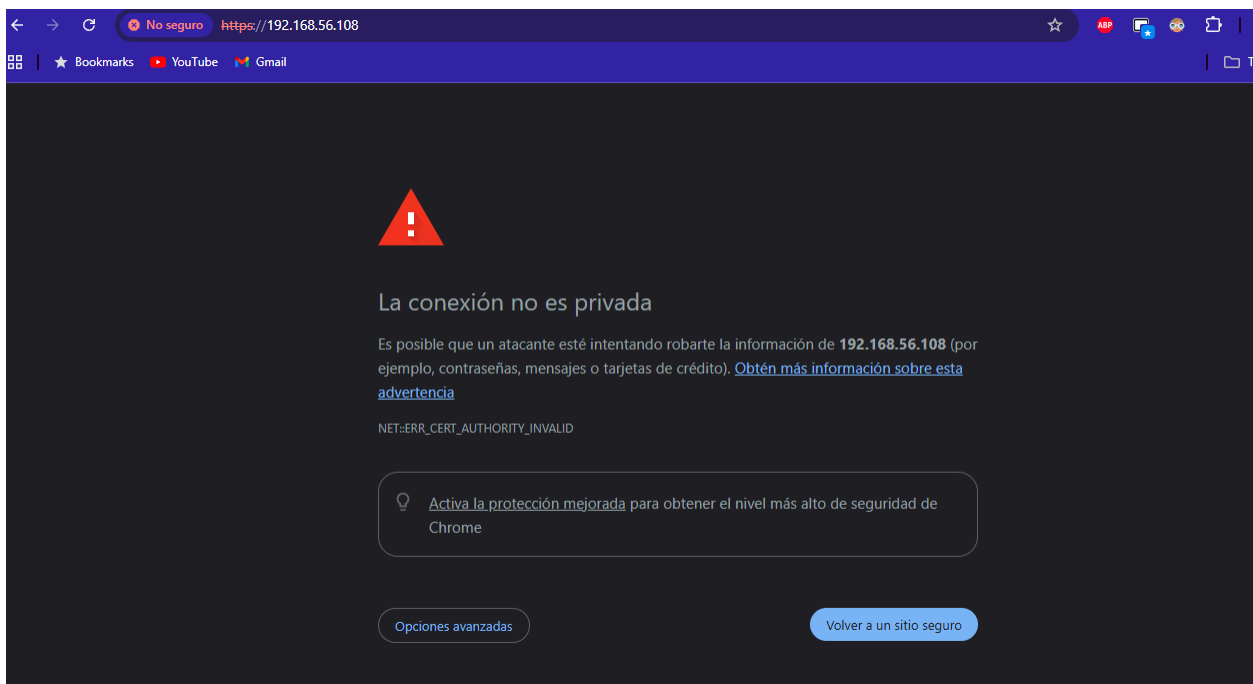
[illegible]



5.3 Configurar sitio HTTPS:

```
mauro@debianhost:~$ sudo cp /etc/apache2/sites-available/default-ssl.conf /etc/apac  
he2/sites-available/misitio-ssl.conf  
sudo nano /etc/apache2/sites-available/misitio-ssl.conf
```

```
mauro@debianhost: ~  
mauro@debianhost:~$ sudo a2ensite misitio-ssl.conf  
sudo systemctl reload apache2  
Enabling site misitio-ssl.  
To activate the new configuration, you need to run:  
systemctl reload apache2  
mauro@debianhost:~$
```



Paso 6: Firewall y puertos



```
mauro@debianhost: ~  
$ sudo ufw enable  
Netid      State      Recv-Q     Send-Q     Peer Address:Port      Local Address:Port  
udp        UNCONN     0           0           0.0.0.0:*          10.0.2.15:68  
udp        UNCONN     0           0           0.0.0.0:*          192.168.56.108:68  
udp        UNCONN     0           0           [fd17:625c:f037:2:40a1:9d53:feee:a048]:546  
[::]:*  
udp        UNCONN     0           0           [fe80::42a9:beaa:cdf3:3bb2]%enp0s3:546  
[::]:*  
udp        UNCONN     0           0           [fe80::46ad:7118:5dee:d805]%enp0s8:546  
[::]:*  
udp        UNCONN     0           0           [fd17:625c:f037:2:a00:27ff:fe4b:113]:546  
[::]:*  
tcp        LISTEN     0           128         0.0.0.0:*          0.0.0.0:22  
tcp        LISTEN     0           4096        0.0.0.0:*          127.0.0.1:37359  
tcp        LISTEN     0           80          0.0.0.0:*          127.0.0.1:3306  
tcp        LISTEN     0           4096        0.0.0.0:*          0.0.0.0:8080  
tcp        LISTEN     0           511         0.0.0.0:*          *:80  
tcp        LISTEN     0           128         [::]:*          [::]:22  
tcp        LISTEN     0           511         [::]:*          *:443  
tcp        LISTEN     0           4096        [::]:*          [::]:8080  
Installing:  
  ufw  
Suggested packages:  
  rsyslog  
Summary:  
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 49  
  Download size: 169 kB  
  Space needed: 880 kB / 16.0 GB available  
Get:1 http://deb.debian.org/debian trixie/main amd64 ufw all 0.36.2-9 [169 kB]  
Fetched 169 kB in 0s (1,085 kB/s)  
Preconfiguring packages ...
```

Paso 7: Revisar logs y monitoreo



```
mauro@debianhost:~$ sudo tail -f /var/log/apache2/access.log
192.168.56.1 - - [20/Mar/2026:01:27:25 -0400] "-" 408 0 "-" "-"
192.168.56.1 - - [20/Mar/2026:01:31:18 -0400] "GET /login.php HTTP/1.1" 200 414 "-"
"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/146.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:01:31:42 -0400] "POST /login.php HTTP/1.1" 200 431 "h
ttp://192.168.56.108/login.php" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWeb
Kit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:01:32:34 -0400] "-" 408 0 "-" "-"
192.168.56.1 - - [20/Mar/2026:01:35:18 -0400] "POST /login.php HTTP/1.1" 200 431 "h
ttp://192.168.56.108/login.php" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWeb
Kit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:01:35:24 -0400] "GET / HTTP/1.1" 200 3383 "-" "Mozill
a/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/1
46.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:01:35:24 -0400] "GET /icons/openlogo-75.png HTTP/1.1"
200 6040 "http://192.168.56.108/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) Apple
WebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:01:36:15 -0400] "-" 408 0 "-" "-"
192.168.56.1 - - [20/Mar/2026:10:32:08 -0400] "GET / HTTP/1.1" 200 3367 "-" "Mozill
a/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/1
46.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:10:33:10 -0400] "-" 408 0 "-" "-"
```

```
mauro@debianhost:~$ sudo tail -f /var/log/apache2/error.log
[Fri Mar 20 01:38:33.767753 2026] [core:notice] [pid 15516:tid 15516] AH00094: Comm
and line: '/usr/sbin/apache2'
[Fri Mar 20 01:39:37.159085 2026] [mpm_prefork:notice] [pid 15516:tid 15516] AH0017
0: caught SIGWINCH, shutting down gracefully
[Fri Mar 20 01:39:37.771894 2026] [mpm_prefork:notice] [pid 15847:tid 15847] AH0016
8: Apache/2.4.66 (Debian) configured -- resuming normal operations
[Fri Mar 20 01:39:37.772020 2026] [core:notice] [pid 15847:tid 15847] AH00094: Comm
and line: '/usr/sbin/apache2'
[Fri Mar 20 10:31:24.665624 2026] [mpm_prefork:notice] [pid 15847:tid 15847] AH0017
1: Graceful restart requested, doing restart
AH00558: apache2: Could not reliably determine the server's fully qualified domain
name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this mes
sage
[Fri Mar 20 10:31:24.798547 2026] [ssl:warn] [pid 15847:tid 15847] AH01906: 127.0.1
1:443:0 server certificate is a CA certificate (BasicConstraints: CA == TRUE !?)
[Fri Mar 20 10:31:24.798608 2026] [ssl:warn] [pid 15847:tid 15847] AH01909: 127.0.1
1:443:0 server certificate does NOT include an ID which matches the server name
[Fri Mar 20 10:31:24.800338 2026] [mpm_prefork:notice] [pid 15847:tid 15847] AH0016
8: Apache/2.4.66 (Debian) OpenSSL/3.5.5 configured -- resuming normal operations
[Fri Mar 20 10:31:24.800375 2026] [core:notice] [pid 15847:tid 15847] AH00094: Comm
and line: '/usr/sbin/apache2'
```

Paso 8: Checklist de Verificación